

3D + 6DOF Mods

Setup & Configuration

Install it, and you already have 3D — it ships **on**. Pick the output for your screen, set your eye distance, and tune it. On a 3D display, or in VR at full resolution. [Page 1 set up](#) · [Page 2 3D displays](#) · [Page 3 VR](#).

What changed in this version. The 3D now ships **on** in Side-by-Side. The eyes alternate cleanly every frame and the picture no longer blinks to mono on a stutter or when you use a game-speed tool. The centre seam is mirrored so it looks clean, and menus / cutscenes stay in your 3D format instead of flipping to flat. There is no longer a 'frame-sequential' shutter mode: full resolution now means VR (mode 6); everything else packs both eyes into one frame.

1 Set up

The mod nudges the game's own camera left and right, one eye per frame; the game redraws the scene from each eye. The depth is real — correct at every distance.

Install

- + Unzip, then copy everything into the folder with the game's .exe. (Steam: right-click → Manage → Browse local files.)
- + Launch the game. If 3D-6DOF .log appears next to it, it's running.

No 3D-6DOF.log? It isn't loading. Check the right .exe folder, run as Administrator once, and check no other mod (ReShade, ENB) uses the same loader name (dinput8.dll).

It's already on — just pick your output

Open 3D-6DOF config.ini in any text editor. It ships enabled in Side-by-Side; set OutputMode for your display (page 2) or VR (page 3).

```
[Stereo3D]
; ---- the 3D ----
Enabled=1          ; ships ON (set 0 for tracking-only, no 3D)
OutputMode=1       ; 1 = Side-by-Side (default) ... 6 = VR headset (page 3). 0 = 3D OFF
SeparationMM=65.0  ; your eye distance in mm
ConvergenceDistance=.. ; where the screen sits (pre-tuned per game; adjust with Ctrl+F5/F6)
SwapEyes=0         ; 1 if the depth looks inside-out
; ---- performance / advanced further down: PresentUncap / MaxFPS / AfrReproject (page 3) ----
```

It's grouped: the 3D first, then performance, then advanced you can ignore. Each line says what it does; this guide says why.

The four knobs

Knob	What it does	INI key	Keys
Separation	3D strength. Your eye distance in mm — most people 65mm; 70mm is a bit more strain.	[SeparationMM] SeparationMM	Ctrl+F3 / F4
Convergence	Where the screen is. Things this far away sit ON it; nearer pops out and sinks in.	[ConvergenceDistance] ConvergenceDistance	Ctrl+F5 / F6
Z offset	Camera closer / back from your character.	[ZAxis] Offset	Ctrl+Down / Up
FOV	A wider view subtly deepens the 3D.	[FOV] Scale	Ctrl+F8 / F9

Two more, any time: **Ctrl+F11** cycles the output mode, **Ctrl+F12** swaps the eyes. Hold Ctrl and tap the key while playing; let go and the value saves itself to the INI. Note the directions: **Ctrl+Down** pulls the camera *closer* to your character, **Ctrl+Up** pushes it *back*.

Getting the numbers right — in order

- + Set SeparationMM to your real IPD (65 if you don't know it).

- + Adjust convergence (Ctrl+F5/F6) until something mid-distance sits on the screen.
- + Then raise separation (Ctrl+F4) for depth. Stop when it feels like work.
- + Depth inside-out? SwapEyes=1 (or tap Ctrl+F12).

Too much separation is the #1 cause of eye strain. Tired after ten minutes = separation too high, not your eyes. Real 3D should feel effortless.

Turn these OFF in the game

One eye is drawn per frame, alternating — so anything that blends frames together mixes your two eyes and destroys the 3D. Turn off **motion blur**, **TAA** (and DLSS / FSR2 / XeSS — use FXAA or SMAA), **frame generation** and **dynamic resolution**.

Head-tracking (optional)

3D and head-tracking are independent. For tracking: OpenTrack → Output = UDP over network → 127.0.0.1:4242 → Start. **Home** recenters, **End** toggles tracking. VR users skip this — the VR viewer does the looking-around itself.

2 3D displays and other output methods

For a 3D TV, 3D monitor, 3D projector or passive glasses. Using a VR headset? Go to page 3. OutputMode picks how the two eyes reach your screen:

Mode	Name	Use when...	Per eye
0	Off	Turns the 3D off (plain 2D). Not a display format.	—
1	Side-by-Side (default)	Your 3D TV / projector has an SBS mode. Most widely supported.	Half width
2	Top-and-Bottom	Your display has a TAB / over-under mode.	Half height
3	Line Interlaced	Passive 3D displays (polarised glasses).	Half height
4	Column Interlaced	Some passive / lenticular displays.	Half width
5	Checkerboard	Older 3D DLP TVs (Samsung / Mitsubishi).	Half
6	VR full-res	A VR headset — page 3.	Full

Which one?

- + **3D TV or projector** → 1, then set the TV's own 3D mode to Side by Side. (Use 2 if it prefers over-under.)
- + **Passive 3D with polarised glasses** → 3.
- + **Old 3D DLP TV** → 5.
- + **VR headset** → 6 (page 3) — the only full-resolution option.
- + Modes 1–5 run at your display's full refresh rate, no flicker, no shutter.

Shutter glasses / 120 Hz 3D: this version has no frame-sequential (alternating-eye) output any more — that path was the source of the old eye-swap/flicker problems. Use Side-by-Side (mode 1) with your display's SBS mode, or a VR headset (mode 6) for full resolution.

Why 1–5 are half resolution (not a bug)

In Side-by-Side, two eyes must fit into one frame — the frame is one screen wide, so each eye gets half the width. Top-and-Bottom splits the height. Interlaced and checkerboard are half because the panel itself gives each eye alternate rows or pixels. That's the format, not the mod — no software can change it. Want full resolution? Use VR (mode 6, page 3).

One eye per frame — and what we do about it

In every packed mode only one real eye is drawn per frame, so each eye updates every other frame. In fast motion the two eyes are one frame apart, which can look like a slight shear. Three things keep it clean:

- + **The eyes alternate exactly 1:1** with the display now — driven by the frame you actually see, not by the game's internal timing — so there's no uneven L/R shimmer, and it stays stable even with game-speed / Cheat-Engine tools running.
- + **The picture never blinks to mono** on an ordinary stutter; it only goes flat during a real camera stop (a menu), and even then it stays in your chosen format.
- + **The centre seam is mirrored**, so the join between the two eyes reads as a clean reflection rather than a smeared strip.
- + For turning, **reprojection** (page 3) can re-aim the older eye — it's off by default because it can add a little judder on analog panning; turn it on if you prefer it.

Tips

- + **Image inside-out?** SwapEyes=1 or tap Ctrl+F12. Common on interlaced displays — it depends which scanline your panel treats as the left eye.
- + **Edges uncomfortable but centre fine?** Lower SeparationMM slightly, or raise ConvergenceDistance.

- + **The HUD will look doubled.** It's drawn flat at screen depth. A nearer ConvergenceDistance reduces it; it can't be fully fixed from a camera mod.
- + **Guardians FOV** is experimental and ships off (its FOV field isn't known yet). Turn on Diagnostics and read the DIAG FOV-candidates log line to find it. Batman's FOV works and is on.

Fixing problems

Problem	Fix
Nothing happens	No 3D-6DOF.log = not loading. Right .exe folder? Run as Admin once. Log names a different game = wrong d
Depth inside-out	SwapEyes=1 (or Ctrl+F12).
Doubled / blurry, no depth	Display isn't in 3D mode, or OutputMode doesn't match it (this page).
Eyes hurt	Lower SeparationMM. Ghosting / smearing = motion blur, TAA or DLSS still on.
Flicker or shimmer	Make sure motion blur / TAA / frame-gen are OFF in-game; those blend the two eyes.
Headset black	OutputMode=6 and Enabled=1? VR viewer running? Log says [vr]?

3 VR — full 3D + 6DOF at native resolution

This is the best mode. Every other mode squeezes two eyes into one screen-sized frame, costing half the resolution. Mode 6 doesn't use the game's frame at all — the mod builds its own double-width image (two full-size eyes side by side), so each eye keeps its native full width. It's also where the framerate options below do the most.

- + A VR viewer that reads the mod's shared stereo image (Katanga / Osiris-class) + a VR headset.
- + OpenTrack not needed — the VR viewer drives the head tracking itself.

Setup

```
[Stereo3D]
Enabled=1
OutputMode=6          ; VR: full-resolution side-by-side shared surface
SeparationMM=65.0     ; your REAL IPD
ConvergenceDistance=.. ; tune with Ctrl+F5/F6
```

- + Start the VR viewer, then launch the game — it picks up the mod's stereo image automatically.
- + Put the headset on. Real 3D, and looking around works with no OpenTrack.

How to tell it's working

- + 3D-6DOF. Log has a line mentioning [vr] or Katanga.
- + Your monitor shows a normal, steady, single picture — that's deliberate. The monitor gets a clean left-eye view; the headset gets the real 3D.

Tuning for VR

- + Use your real IPD — guessing high is what causes discomfort.
- + Convergence (Ctrl+F5/F6) sets how far the world feels; Z offset (Ctrl+Down/Up) pulls the camera toward / away from your character.

Honest limits. This moves the game's own camera, one eye per frame — real 3D in games with no VR support, but not a native VR port. Not roomscale; the HUD is flat.

Performance — the settings that matter

Per-eye fps = game fps ÷ 2, so the only way to raise it is to let the game run faster. In VR the mod does this for you: the headset reads the mod's image, so the desktop window isn't the output, and the mod stops it being vsynced to your monitor, which was throttling the game for a window nobody's looking at.

Setting	What to use
PresentUncap	1 = VR only (default, free). 2 = also the 3D-display modes: 2x per eye, cost is tearing. 0 = off.
MaxFPS	0 = unlimited, or 2x your refresh (180 for a 90 Hz headset, 120 for a 60 Hz 3D TV) - a full fresh image per eye, GPU handled.
AfrReproject	0 = off (default). 1 re-aims the eye that wasn't redrawn toward where you're looking, so both eyes agree on turns - but can't be perfect.
In-game vsync	Off, and turn off the game's own fps cap. The mod can't override those - if the game holds itself to 60 you're stuck at 30 per eye.
Upscaling / AA	DLSS, FSR2, XeSS, frame-gen and TAA can't work here - they blend the previous frame, which is the other eye. Use FXAA.

These two mods are alternate-frame (AFR) 3D: both eyes are real renders, one frame apart. There is no depth-reconstruction (DIBR) mode here — every pixel of each eye is genuinely drawn by the game.

3D + 6DOF Mods — Setup & Configuration. AFR stereo, four independent layers (3D, tracking, FOV, Z-dolly), full-resolution VR via mode 6. Companion to the technical method document.